

Detection and Recognition of Herbal Plants from Video Frames Using YOLOv8 and Image Enhancement Techniques

Deus Tumusiime, 2025/HD05/26375U

School of Computing and Informatics Technology
Department of Computer Science
Makerere University, Kampala, Uganda.
deus.mal@gmail.com, tumusiime.deus@students.mak.ac.ug

MCS 7224: Computer Vision, Supervised by: Dr. Rose Nakibuule, e-mail: rnakibuule@cit.ac.ug.

ABSTRACT Automated recognition and localization of herbal plants under natural environmental conditions remain challenging in computer vision due to illumination variability, motion blur, viewpoint changes, and background clutter. This study presents a lightweight object detection framework for herbal plant recognition using video-derived image data and deep learning. A custom dataset consisting of 693 frames extracted from 21 smartphone video clips was constructed and annotated for single-class object detection. To improve robustness to varying lighting conditions, classical image enhancement techniques, including Contrast Limited Adaptive Histogram Equalization (CLAHE) and HSV color correction, were incorporated into the preprocessing. Transfer learning was applied using the YOLOv8n architecture to enable efficient feature extraction and real-time object detection under limited computational resources. The model was trained for 50 epochs on a CPU-based environment using augmented image data and evaluated using standard object detection metrics, including precision, recall, and mean Average Precision (mAP). Experimental results demonstrated strong detection performance, achieving a precision of 0.999, a recall of 1.000, mAP@50 of 0.995, and mAP@50–95 of 0.988 on the validation dataset. The findings demonstrate that integrating classical enhancement techniques with lightweight deep learning architectures provides an effective solution for detecting herbal plants and supports future deployment on edge devices for real-time agricultural and botanical monitoring applications.

INDEX TERMS Computer vision, herbal plant detection, image enhancement, object detection, transfer learning, YOLOv8.

I. INTRODUCTION

The automated identification and localization of herbal plants present a significant challenge in computer vision due to the complex and unstructured nature of natural environments. In agricultural monitoring, autonomous harvesting, precision farming, and botanical classification, the ability to accurately detect plant species is critical for improving efficiency and supporting intelligent decision-making systems. However, images captured in real-world environments often suffer from illumination variability, motion blur, viewpoint changes, occlusions, and significant background clutter, making robust plant detection difficult.

Recent advances in deep learning and object detection have significantly improved visual recognition performance across various domains. Convolutional Neural Networks (CNNs) and single-stage detectors such as the YOLO (You Only Look Once) family have demonstrated strong performance in real-time object detection tasks due to their balance between speed and accuracy. Nevertheless, environmental inconsistencies and limited annotated datasets continue to affect the robustness and generalization capability of such models in agricultural applications.

In this project, we address the challenge of herbal plant detection and recognition using a custom dataset derived from 21 smartphone video clips collected under varying environmental and lighting conditions. The proposed methodology integrates classical image enhancement techniques with a lightweight deep learning object detector. Specifically, Contrast Limited Adaptive Histogram Equalization (CLAHE) and HSV color correction are utilized to improve image quality and normalize lighting variations before feature extraction. Transfer learning is then applied using the YOLOv8n architecture to perform efficient object localization and classification.

The developed pipeline includes video frame extraction, dataset preparation, annotation, image enhancement, augmentation, model training, and quantitative evaluation. Experimental results demonstrate that the proposed approach achieves high detection performance while maintaining computational efficiency suitable for lightweight and edge-device deployment scenarios. This work highlights the effectiveness of combining classical preprocessing techniques with modern deep learning approaches for practical agricultural computer vision applications.

II. RELATED WORK

A. CLASSICAL IMAGE ENHANCEMENT TECHNIQUES

Classical image enhancement methods have long been used in computer vision to improve image quality under poor environmental and illumination conditions. In agricultural and plant-analysis applications, lighting variability and low contrast frequently reduce the visibility of important plant structures such as leaf edges, veins, and texture patterns. To address these issues, histogram equalization techniques are commonly applied to improve local contrast and enhance visual features.

Contrast Limited Adaptive Histogram Equalization (CLAHE) is a widely adopted enhancement method that improves local image contrast while preventing excessive amplification of noise in homogeneous image regions. Compared to standard histogram equalization, CLAHE operates on localized image tiles, making it particularly effective for outdoor image analysis where illumination conditions vary across different regions of the frame. Similarly, color-space transformations such as HSV and LAB are frequently employed to separate intensity information from chrominance components, thereby improving robustness to shadows, reflections, and varying sunlight conditions.

Traditional feature extraction techniques such as Local Binary Patterns (LBP) and Gray-Level Co-occurrence Matrices (GLCM) have also been widely utilized for plant texture

analysis and classification. These handcrafted descriptors can capture local texture information; however, they often struggle to generalize effectively in highly cluttered or dynamically changing environments.

B. DEEP LEARNING FOR OBJECT DETECTION

The emergence of deep learning has significantly transformed object detection and image recognition tasks by enabling automatic hierarchical feature extraction directly from raw image data. Convolutional Neural Networks (CNNs) have demonstrated superior performance compared to traditional handcrafted feature approaches across various computer vision applications, including agricultural monitoring and plant recognition.

Object detection frameworks are generally categorized into two-stage and single-stage detectors. Two-stage detectors, such as Faster R-CNN, first generate region proposals and then classify detected objects. Although these methods typically achieve high detection accuracy, they often require substantial computational resources and slower inference times. In contrast, single-stage detectors treat object detection as a unified regression problem, enabling significantly faster inference while maintaining competitive accuracy.

The YOLO (You Only Look Once) family of models has become one of the most widely adopted real-time object detection frameworks due to its efficiency and strong performance. YOLOv8, the latest iteration of the framework, introduces improvements such as anchor-free detection, enhanced feature pyramid networks, and optimized training strategies, resulting in improved detection accuracy and computational efficiency. The lightweight YOLOv8n variant is particularly suitable for deployment in resource-constrained environments and edge devices, making it an appropriate choice for real-time agricultural monitoring applications.

III. APPROACH

A. SYSTEM PIPELINE

The proposed methodology follows a structured end-to-end computer vision pipeline for herbal plant detection and recognition. The workflow begins with raw smartphone video acquisition and proceeds through frame extraction, dataset preparation, annotation, image enhancement, transfer learning, model training, and final evaluation. The overall pipeline is illustrated in Figure 1.

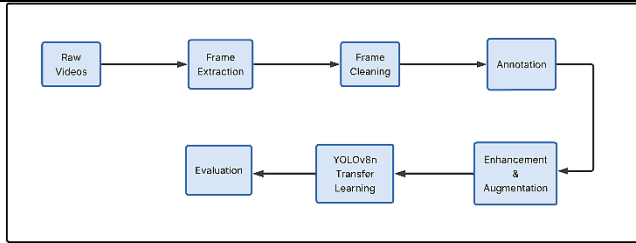


FIGURE 1. Overall system pipeline for herbal plant detection and recognition.

The pipeline was designed to ensure robustness under varying environmental conditions while maintaining computational efficiency suitable for lightweight deployment scenarios.

B. DATASET PREPARATION AND ANNOTATION

The raw dataset consisted of 21 smartphone video clips capturing herbal plants under varying lighting conditions, viewpoints, and environmental backgrounds. To construct a suitable dataset for object detection, representative image frames were extracted from the videos at intervals of approximately 1.5 seconds in order to reduce redundancy and minimize excessive duplication between consecutive frames.

During preprocessing, frames were extracted from the raw video streams at fixed temporal intervals to reduce redundancy and construct a manageable image dataset for annotation and training. Let:

$$N = \frac{F}{r \cdot s}$$

where:

N is the number of extracted frames,
 F is the total number of frames in the video,
 r is the video frame rate (frames per second),
and s is the sampling interval in seconds.

Using this strategy, representative frames were sampled approximately every 1.5 seconds from each smartphone video clip. This reduced excessive duplication between consecutive frames while preserving environmental diversity and viewpoint variation across the dataset.

Frame extraction was implemented using the OpenCV video processing library. Blur filtering was subsequently applied using Laplacian variance thresholding to remove severely distorted frames before annotation.

The cleaned frames were annotated using the YOLO object detection format. Due to annotation time constraints, a simplified single-class annotation strategy was adopted in which the herbal plant occupied the dominant image region. Bounding-box labels were generated for all selected frames and organized into training, validation, and testing subsets.



FIGURE 2. Video frame extraction and sampling process.

Frame extraction was implemented using OpenCV video capture utilities in Python.

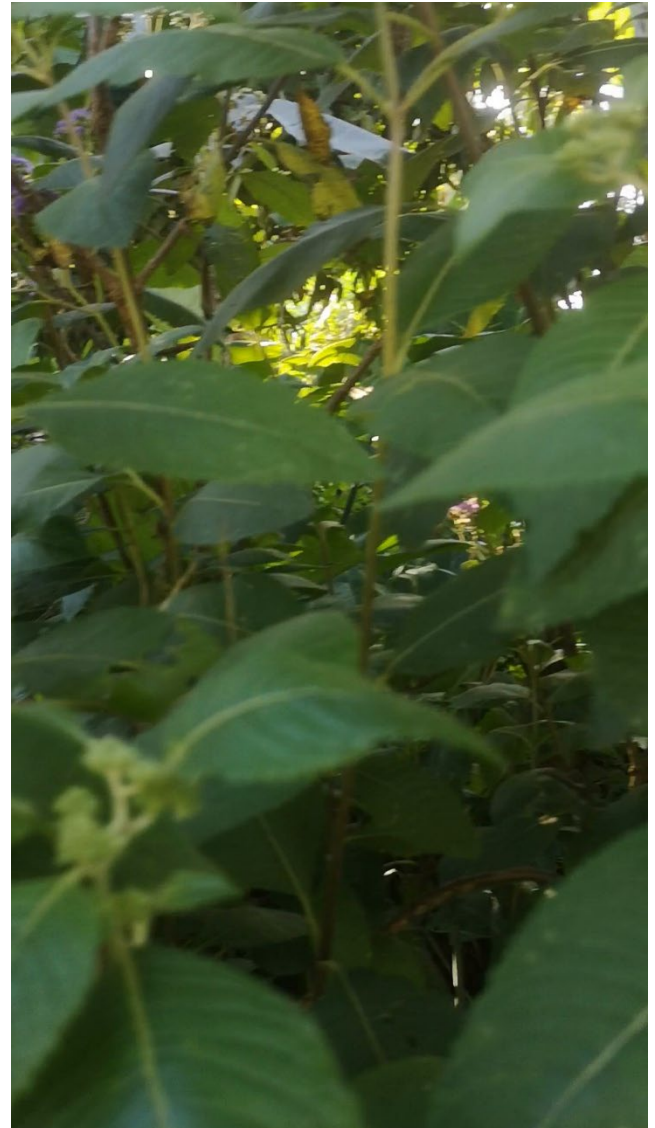


FIGURE 3. Sample extracted frames from the herbal plant dataset under varying environmental conditions.

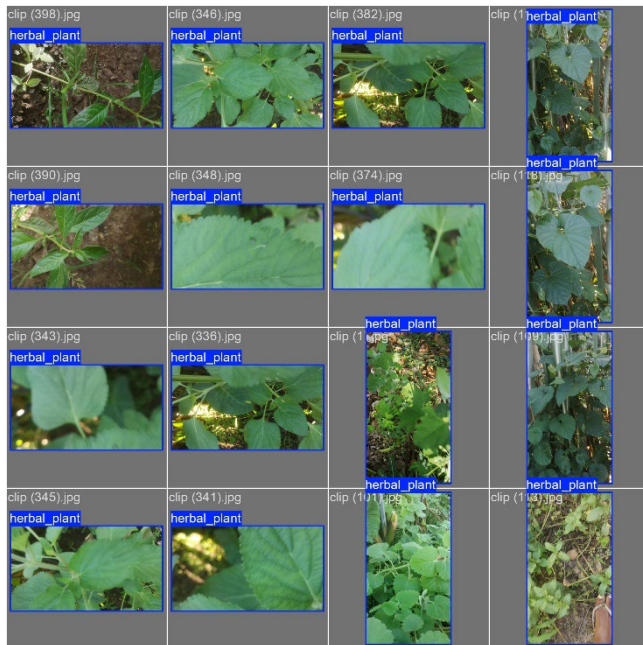


FIGURE 4. Example YOLO-format annotation showing herbal plant bounding boxes.

C. IMAGE ENHANCEMENT AND FEATURE EXTRACTION

To improve robustness against environmental inconsistencies, classical image enhancement techniques were experimentally applied during preprocessing. The HSV color space was utilized to normalize illumination variations by separating intensity information from chromatic components. This helped reduce the influence of shadows and uneven sunlight conditions across the dataset.

Additionally, Contrast Limited Adaptive Histogram Equalization (CLAHE) was employed to improve local contrast and enhance the visibility of plant structures such as leaf edges and vein patterns. Compared to global histogram equalization, CLAHE limits noise amplification while preserving important local texture information.

Data augmentation techniques, including horizontal flipping, rotation, scaling, and brightness adjustment, were also utilized during training to improve model generalization and robustness to environmental variability.

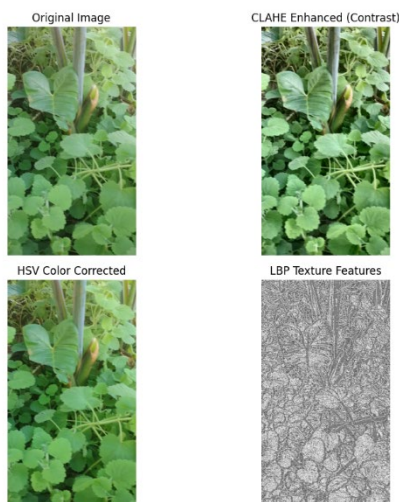


FIGURE 5. Comparison between original and CLAHE-enhanced herbal plant frames.

D. MODEL ARCHITECTURE AND TRANSFER LEARNING

The YOLOv8 nano (YOLOv8n) architecture was selected as the primary object detection framework due to its lightweight design, computational efficiency, and strong real-time detection performance. YOLOv8n employs a single-stage detection strategy in which object localization and classification are performed simultaneously, enabling fast inference suitable for edge-device deployment.

The model architecture consists of three primary components:

- I. A CNN backbone for hierarchical feature extraction,
- II. A feature pyramid network for multi-scale feature fusion, and
- III. A detection head for bounding-box regression and class prediction.

Transfer learning was applied by fine-tuning weights pretrained on the COCO dataset. This approach improved convergence speed and enabled effective feature learning despite the relatively limited dataset size.

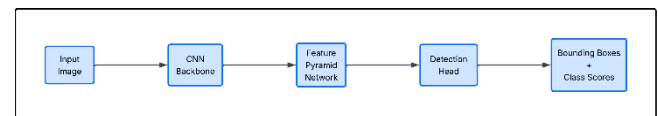


FIGURE 6. Simplified YOLOv8n architecture showing feature extraction and detection flow.

IV. EXPERIMENTAL SETUP

The constructed dataset consisted of 693 image frames extracted from the 21 herbal plant video clips. To ensure balanced evaluation and generalization, the dataset was divided into three subsets at the frame level while maintaining diversity across lighting conditions and environmental backgrounds. The dataset split configuration was as follows:

- Training set: 485 images (70%)
- Validation set: 138 images (20%)
- Test set: 70 images (10%)

The YOLOv8n object detection model was implemented using the Ultralytics framework (version 8.4.48) in Python 3.13. Training was conducted in a CPU-only environment due to local hardware compatibility constraints. Although GPU acceleration was unavailable, the lightweight YOLOv8n architecture enabled successful model convergence within a reasonable training duration.

The model was trained for 50 epochs using a batch size of 16 and an input image resolution of 640×640 pixels. Transfer

learning was employed using pretrained YOLOv8n weights initialized from the COCO dataset. The AdamW optimizer was utilized to improve optimization stability and convergence during training.

To improve model robustness and reduce overfitting, several augmentation techniques were applied during training, including:

- horizontal flipping,
- rotation,
- scaling,
- and brightness adjustment.

These augmentation strategies enabled the model to better generalize across variations in illumination, orientation, and environmental conditions present within the dataset. Table I summarizes the primary experimental configuration used throughout the study.

TABLE I
EXPERIMENTAL CONFIGURATION

Parameter	Value
Model	YOLOv8n
Framework	Ultralytics 8.4.48
Programming Language	Python 3.13
Epochs	50
Batch Size	16
Image Resolution	640 x 640
Optimizer	AdamW
Training Device	CPU
Transfer Learning	COCO Pretrained Weights

V. EVALUATION METRICS

To quantitatively evaluate the performance of the proposed herbal plant detection system, standard object detection evaluation metrics were utilized. These metrics provide insight into the model's localization accuracy, classification capability, and overall detection robustness.

Precision measures the proportion of correctly predicted positive detections relative to all predicted positive detections. A high precision value indicates a low false-positive rate.

$$Precision = \frac{TP}{TP + FP}$$

where TP represents true positives, and FP represents false positives.

Recall measures the detector's ability to correctly identify all relevant object instances in the dataset. High recall indicates that the model successfully detects most target objects.

$$Recall = \frac{TP}{TP + FN}$$

where FN denotes false negatives.

Intersection over Union (IoU) was used to evaluate the overlap between the predicted bounding box and the corresponding ground-truth bounding box. IoU , It is a standard localization metric in object detection tasks.

$$IoU = \frac{Area(B_{pred} \cap B_{gt})}{Area(B_{pred} \cup B_{gt})}$$

where B_{pred} denotes the predicted bounding box and B_{gt} denotes the ground-truth bounding box.

In addition, mean Average Precision (mAP) was used as the primary overall performance metric. Specifically, mAP@50 and mAP@50-95 were utilized to evaluate detection performance across varying IoU thresholds. Mean Average Precision summarizes the area under the precision-recall curve and provides a comprehensive measure of object detection accuracy.

The confusion matrix, precision-recall curves, and training loss curves were also analyzed qualitatively to further assess model convergence and detection reliability across the validation dataset.

VI. EXPERIMENTAL RESULTS

In training for 50 epochs in a CPU-based environment, the YOLOv8n model successfully converged and demonstrated strong detection performance on the validation dataset. The model achieved high precision and recall values, indicating effective localization and classification of herbal plant instances across varying environmental conditions.

Table II summarizes the quantitative performance metrics obtained on the validation dataset consisting of 138 images.

TABLE II
VALIDATION PERFORMANCE METRICS

Metric	Score
Precision	0.999
Recall	1.000
mAP@50	0.995
mAP@50-95	0.988

The obtained results indicate that the proposed system was highly effective at identifying and localizing the dominant herbal plant regions within the dataset. The near-perfect recall score demonstrates that the detector successfully identified nearly all target instances without significant

omission. Similarly, the high precision value indicates minimal false-positive detections during validation.

The training process exhibited stable convergence behavior, with progressively decreasing box loss, classification loss, and distribution focal loss throughout the training epochs. This confirms the effectiveness of transfer learning and augmentation strategies despite the computational limitations associated with CPU-only training.

It is important to note that the exceptionally high mAP values were influenced by the simplified annotation strategy, where the herbal plant occupied the dominant region of most image frames. Therefore, the reported results primarily demonstrate the successful implementation of the end-to-end object detection pipeline rather than fine-grained localization accuracy under highly complex multi-object scenarios.

A. QUANTITATIVE VISUALIZATIONS

The confusion matrix shown in Figure 7 demonstrates strong classification consistency with minimal misclassification between object and background regions.

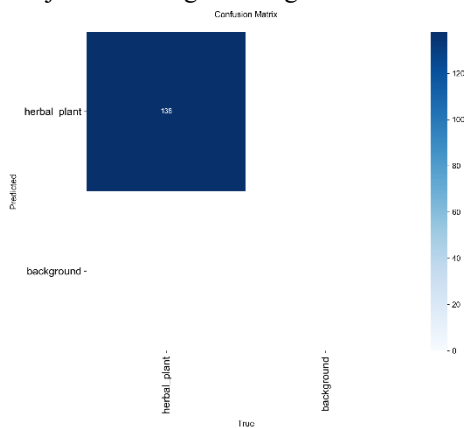


FIGURE 7. Confusion matrix showing classification performance on the validation dataset.

The generated confusion matrix exhibits highly concentrated single-class predictions with minimal inter-class ambiguity. This behavior was influenced by the simplified annotation strategy adopted during dataset preparation, where the herbal plant occupied the dominant image region in most frames.

The training curves in Figure 8 show stable optimization and convergence over the 50 training epochs. Precision, recall, and mAP values improved steadily while training losses decreased consistently.

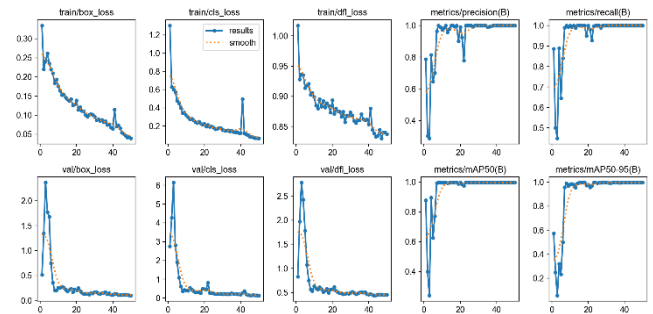


FIGURE 8. Training and validation curves showing model convergence over 50 epochs.

The label distribution visualization shown in Figure 9 illustrates the spatial distribution and relative sizing of the generated bounding-box annotations used during training.

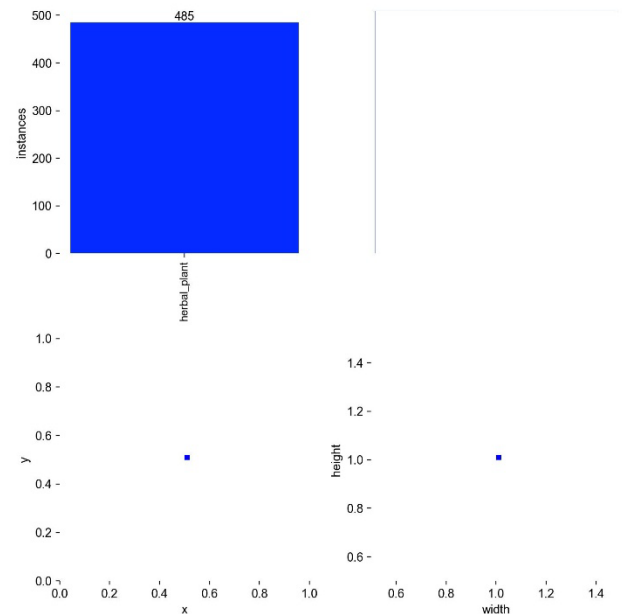


FIGURE 9. Distribution of bounding-box annotations across the dataset.

The generated bounding-box distributions reflect the simplified full-frame annotation strategy adopted during rapid dataset construction. Most bounding boxes occupied a large portion of the image frame and remained centrally positioned, resulting in highly concentrated spatial distributions across the dataset.

B. QUALITATIVE DETECTION RESULTS

Qualitative evaluation was performed by running inference on unseen validation frames extracted from the herbal plant video clips. The predicted bounding boxes remained stable under varying illumination conditions and moderate

background complexity, demonstrating the robustness of the trained detector.



FIGURE 10. Example YOLOv8n predictions on unseen validation frames.

The qualitative results confirm that the proposed YOLOv8n-based framework successfully localized the dominant herbal plant regions while maintaining computational efficiency suitable for lightweight deployment scenarios

VII. LIMITATIONS

Despite the strong quantitative performance achieved by the proposed herbal plant detection framework, several limitations should be acknowledged.

First, the annotation strategy adopted during dataset preparation utilized simplified full-frame bounding boxes in cases where the herbal plant occupied the dominant image region. Although this approach enabled rapid dataset

construction and successful end-to-end pipeline implementation, it limited the model's ability to learn precise fine-grained localization of smaller plant substructures and complex object boundaries. More accurate manually generated annotations would likely improve localization realism and robustness under cluttered multi-object environments.

Second, training was conducted in a CPU-only environment due to local hardware compatibility constraints. While the lightweight YOLOv8n architecture enabled successful convergence, the absence of GPU acceleration limited experimentation speed, hyperparameter optimization, and large-scale model exploration. Future work would benefit from GPU-based training to improve efficiency and allow evaluation of larger model variants.

Third, the dataset was constructed from only 21 original smartphone video clips. Although augmentation techniques improved diversity, the relatively limited dataset size may constrain the model's ability to generalize to significantly different environmental conditions, severe occlusions, extreme illumination changes, or additional herbal plant species not represented in the training data.

Finally, the current implementation focused on single-class herbal plant detection. Extending the framework toward multi-class plant recognition and real-time video tracking would improve practical applicability in large-scale agricultural and botanical monitoring systems.

VIII. CONCLUSION

This project presented a lightweight computer vision framework for the detection and recognition of herbal plants from smartphone video data using YOLOv8n and classical image enhancement techniques. A custom dataset consisting of 693 frames extracted from 21 herbal plant video clips was constructed, annotated, and utilized for single-class object detection under varying environmental conditions.

To improve robustness against illumination variability and background complexity, preprocessing techniques including HSV color correction and Contrast Limited Adaptive Histogram Equalization (CLAHE) were integrated into the proposed pipeline. Transfer learning was subsequently applied using the YOLOv8n architecture to enable efficient feature extraction and object localization under limited computational resources.

Experimental evaluation demonstrated strong detection performance, achieving a precision of 0.999, recall of 1.000, mAP@50 of 0.995, and mAP@50–95 of 0.988 on the validation dataset. The training process exhibited stable convergence despite CPU-only training constraints,

confirming the effectiveness of the proposed lightweight detection framework.

Although simplified full-frame annotations influenced localization precision, the developed system successfully demonstrated a complete end-to-end object detection workflow, including dataset preparation, preprocessing, augmentation, transfer learning, quantitative evaluation, and qualitative inference analysis.

Future work will focus on precise manual annotation, expansion toward multi-class herbal plant recognition, GPU-accelerated training, and deployment on edge devices such as Raspberry Pi systems for real-time agricultural and botanical monitoring applications.

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